

A2A Agent Lab — Prerequisites

Clone the repository



<https://github.com/ehsavoie/a2a-agent-lab-2026>

Scan the QR code or visit the link above to access the repository.

```
git clone https://github.com/ehsavoie/a2a-agent-lab-2026
cd a2a-agent-lab-2026
```

Required software

Tool	Version	Purpose
JDK	21+	Java development
Maven	3.9+	Build tool
Python	3.11+	Exercise 3 (Travel Agent)
Podman / Docker	any	PostgreSQL and Kafka for Exercise 6 (Enterprise)
OpenAI API key	API billing enabled	Java LLM exercises
curl	any	API testing
IDE	IntelliJ / VS Code / etc.	Code editor

1. Set up your API key

The Java agent examples call the OpenAI API. Set `OPENAI_API_KEY` to an API key with API billing enabled. A ChatGPT subscription does not cover API usage, and the GPT-6 Luna API Free tier is unsupported.

```
export OPENAI_API_KEY=your-api-key-here
```

2. Start the Exercise 6 infrastructure (bonus exercise only)

Exercise 6 (Load-Balanced A2A) requires PostgreSQL and Kafka to share state across agent instances. Use Podman Compose or Docker Compose from the exercise directory:

```
cd exercises/exercise-6-enterprise
podman-compose up -d
# or: docker compose -f podman-compose.yml up -d
```

This starts:

- **PostgreSQL 17** on port 5432 — JPA-backed task store shared across WildFly nodes
- **Kafka 4.1** on port 9092 — replicated queue manager for cross-node task events

Exercises 1–5 do not require Docker or Podman.

3. Project structure

a2a-agent-lab-2026/	
pom.xml	← Root aggregator
conference-data/sessions.json	← Shared conference schedule data
docs/	← Step-by-step exercise guides
exercises/	
exercise-1-schedule-advisor/	← Quarkus + A2A Java SDK
exercise-2-venue-agent/	← Spring Boot + spring-a2a
exercise-3-travel-agent/	← Python A2A SDK
travel_agent.py	← Agent implementation
java-client/	← Java interop test client
exercise-4-expense-agent/	← WildFly 41 + Jakarta EE
expense-agent/	← A2A agent (LangChain4j + compliance)
expense-client/	← JAX-RS A2A client
exercise-5-orchestrator/	← Multi-project Quarkus Orchestrator
concierge/	← Multi-tenant: Schedule + Venue + Travel + Expense
orchestrator/	← @SupervisorAgent + @A2AClientAgent + REST UI
exercise-6-enterprise/	← Multi-node WildFly + JPA + Kafka
node-a/ node-b/	← WildFly agent nodes
server/	← Shared server module
client/	← Cross-node test client
initdb/	← PostgreSQL seed scripts
podman-compose.yml	← PostgreSQL + Kafka infrastructure
solutions/	← Complete reference solutions